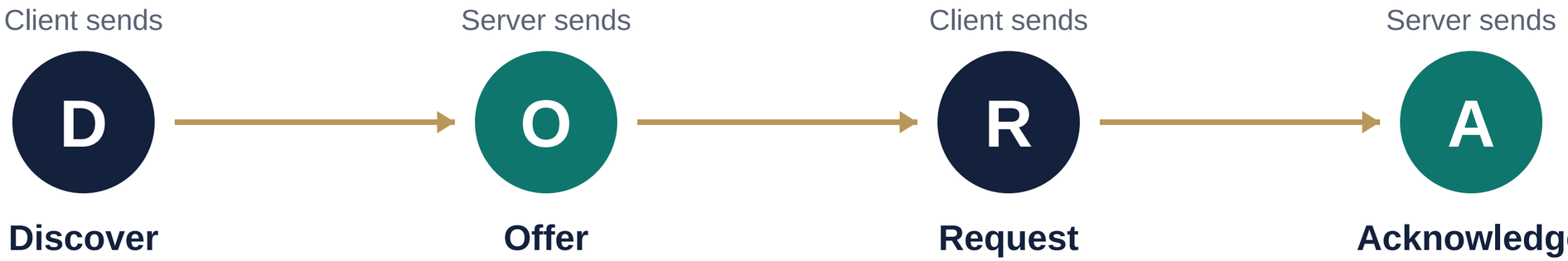
Mawlana Bhashani Science and Technology University
Department of Information and Communication Technology

DHCP

Dynamic Host Configuration Protocol

Data Communication Lab · ICT-3102



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OVERVIEW

Agenda

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Basics

The problem and what DHCP is

2

How it works

Ports, DORA and leases

3

In practice

Allocation, relay agents and messages

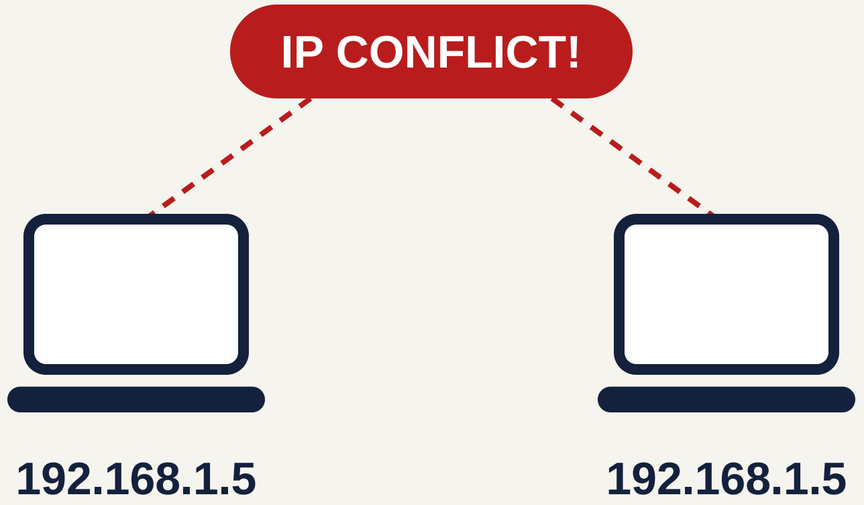
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Trade-offs and security

Advantages, attacks and comparisons

The problem: manual IP configuration

- Every device needs a unique IP, mask, gateway and DNS
- Typing them by hand for hundreds of devices is slow
- One typo creates a duplicate IP and a conflict
- Moving a device to another network means redoing it



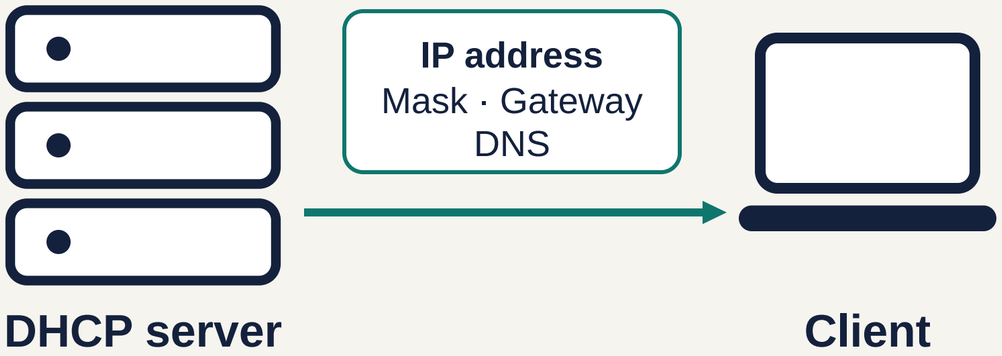
500 PCs × 4 settings

= 2,000 manual entries

What is DHCP?

DHCP is an application-layer client-server protocol that automatically assigns IP configuration to hosts joining a network.

A central server replaces manual setup on every device.



Settings arrive automatically

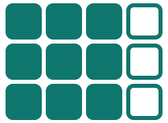
BASICS

What DHCP assigns



IP address

Unique identity, leased for a limited time



Subnet mask

Shows the size of the local network



Gateway

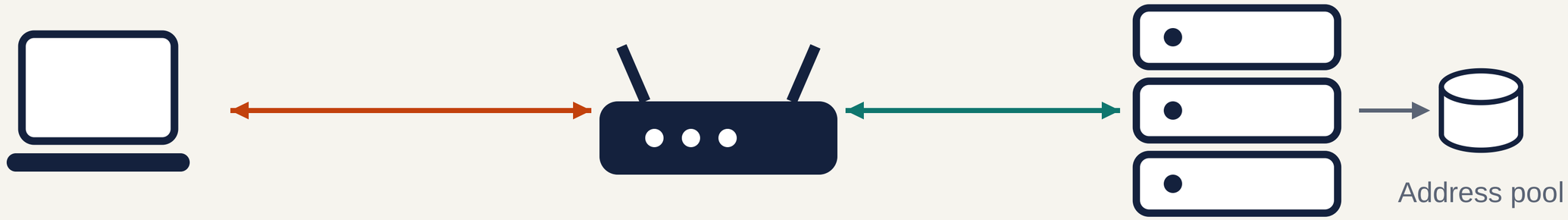
Router used to leave the network



DNS server

Turns names into IP addresses

Client-server architecture



Client
Laptop, phone, printer

Relay agent
Forwards to another subnet

DHCP server
Pool and lease records

A relay agent is needed only when the server is on another network.

Ports and transport



Why UDP? The client has no IP address yet, so it cannot open a TCP connection. UDP lets it simply broadcast.

HOW IT WORKS

The DORA process

D

Discover

Client sends

Client broadcasts:
any DHCP server here?

O

Offer

Server sends

Server proposes
an IP address

R

Request

Client sends

Client accepts
one offer

A

Acknowledge

Server sends

Server confirms
the lease

Discover and Offer



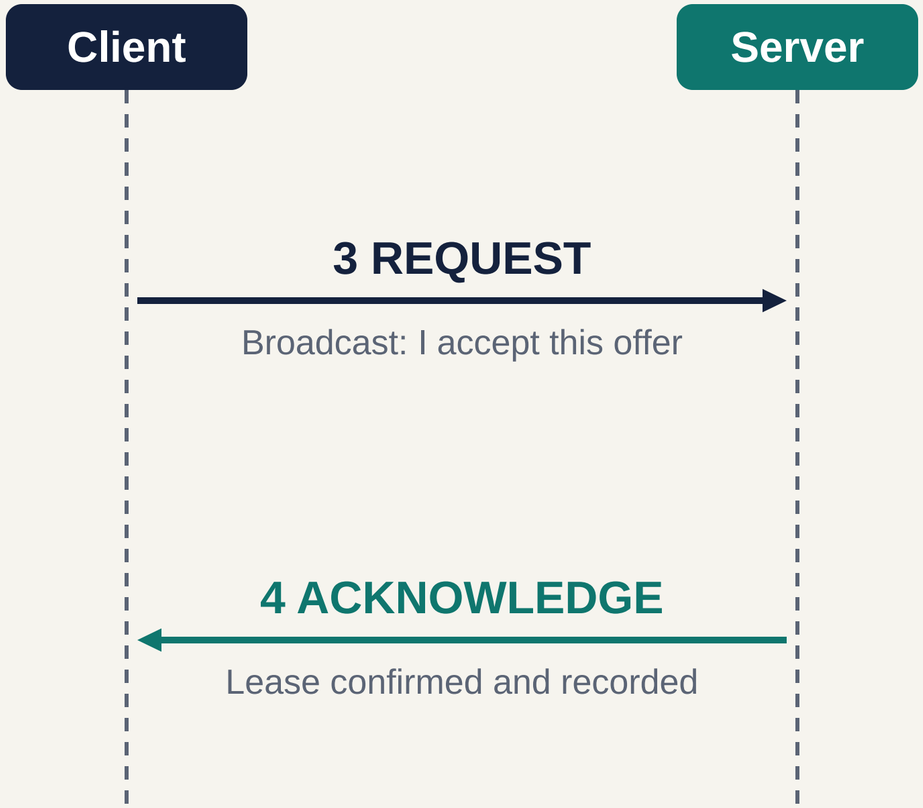
Discover

The client broadcasts from 0.0.0.0 because it has no address yet.

Offer

The server replies with a free IP address and a lease time.

Request and Acknowledge



Request

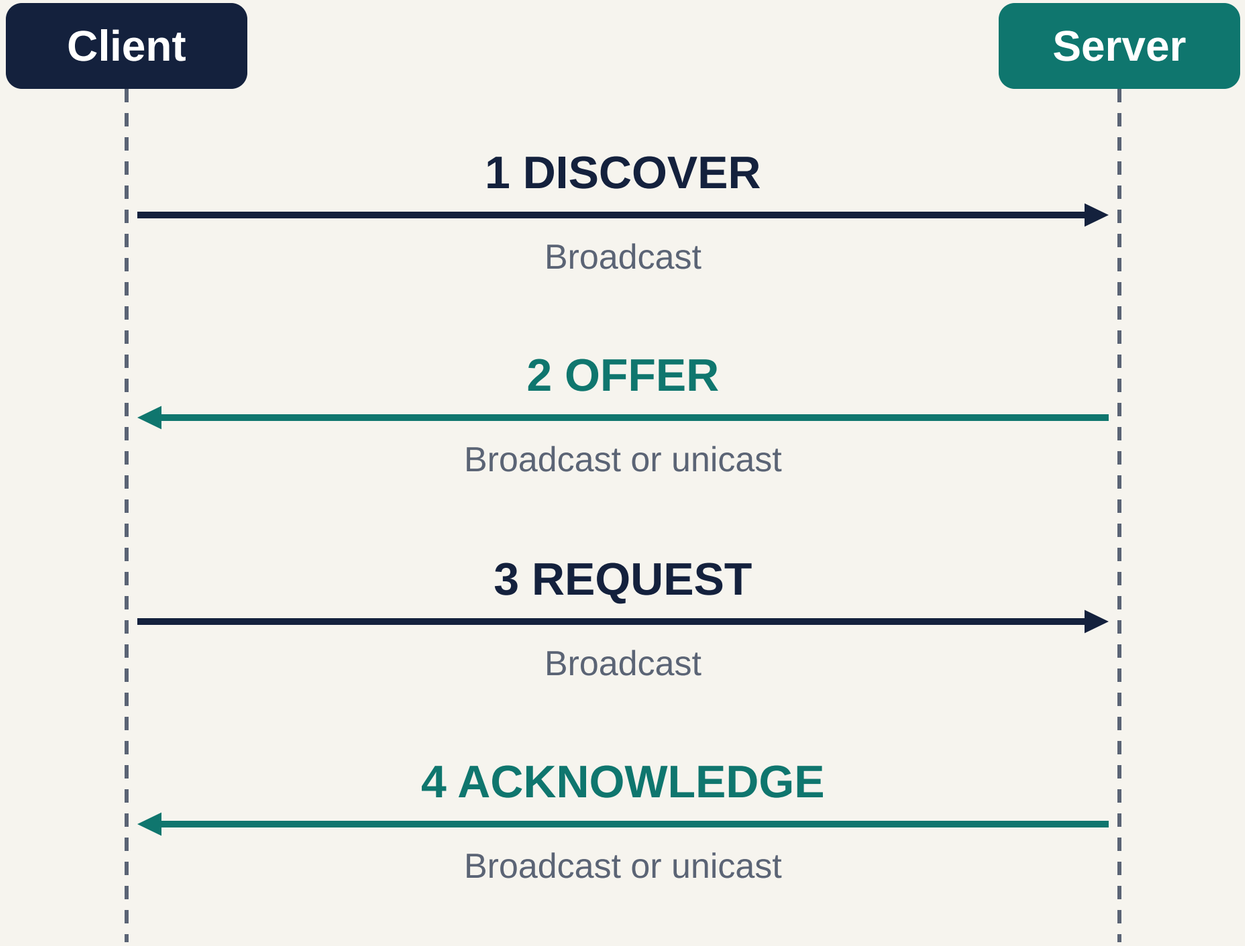
The client broadcasts its choice, so other servers drop their offers.

Acknowledge

The server confirms the lease and the client configures itself.

HOW IT WORKS

DORA packet flow



Client sends

Steps 1 and 3

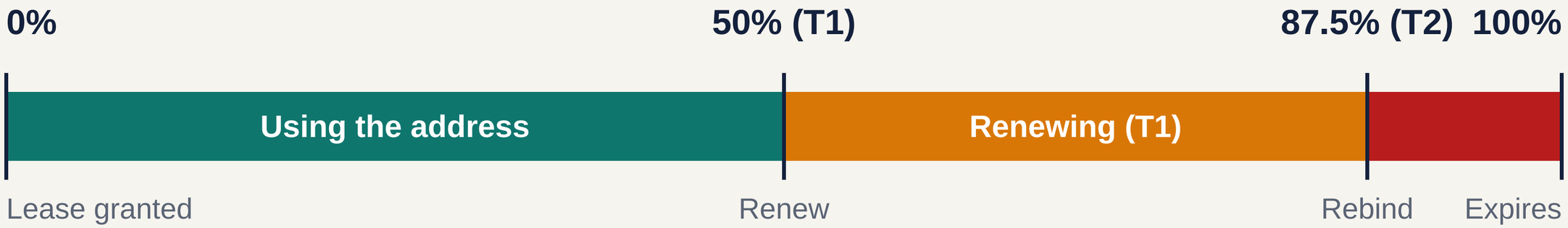
Server sends

Steps 2 and 4

Broadcast vs unicast

Broadcast goes to all hosts. Unicast goes to one host.

Lease time and renewal



T1 at 50%

The client asks the same server to renew the lease.

T2 at 87.5%

The client asks any server to rebind.

100%

The lease expires. The address returns to the pool.

IN PRACTICE

Allocation types



Dynamic

Address from the pool for a limited lease.

Best for: Laptops, phones



Automatic

Address from the pool, kept for a long time.

Best for: Stable hosts



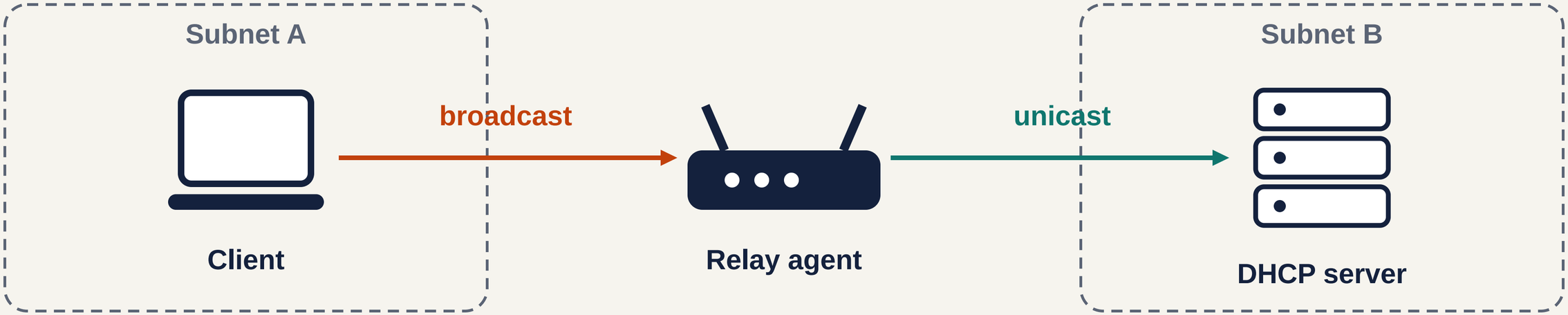
Manual

Fixed IP tied to a MAC address (reservation).

Best for: Servers, printers

IN PRACTICE

DHCP relay agent



- Routers do not forward broadcasts by default
- The relay catches the client broadcast and sends it as unicast
- One central server can then serve many subnets

IN PRACTICE

Other DHCP messages

Sent by client

Decline

The offered address is already in use.

Sent by server

NAK

Request refused, so the client restarts DORA.

Sent by client

Release

Gives the address back before the lease ends.

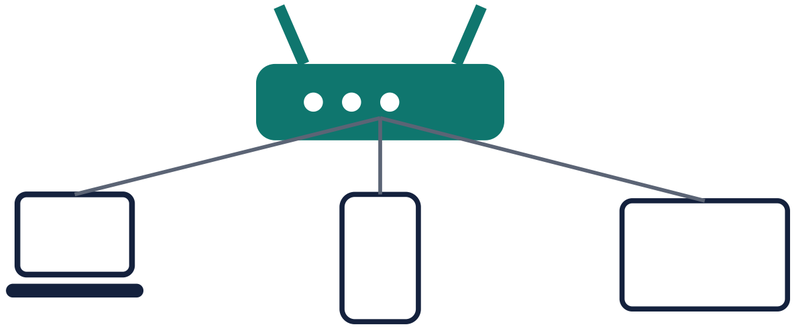
Sent by client

Inform

Already has an IP and asks only for extra settings.

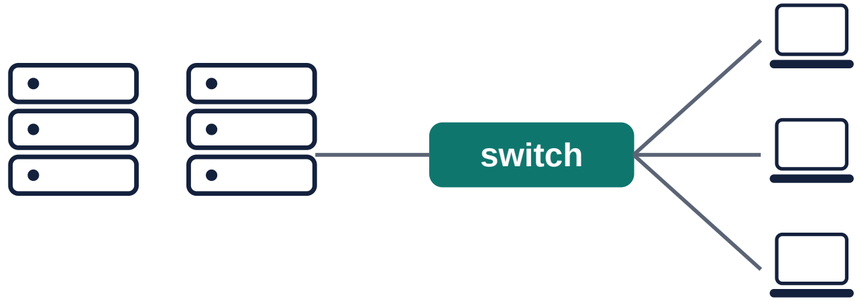
IN PRACTICE

Home router vs enterprise



Home router

Built-in server, one small pool, defaults rarely changed.



Enterprise

Dedicated servers, many scopes and VLANs, redundancy and reservations.

Advantages and disadvantages

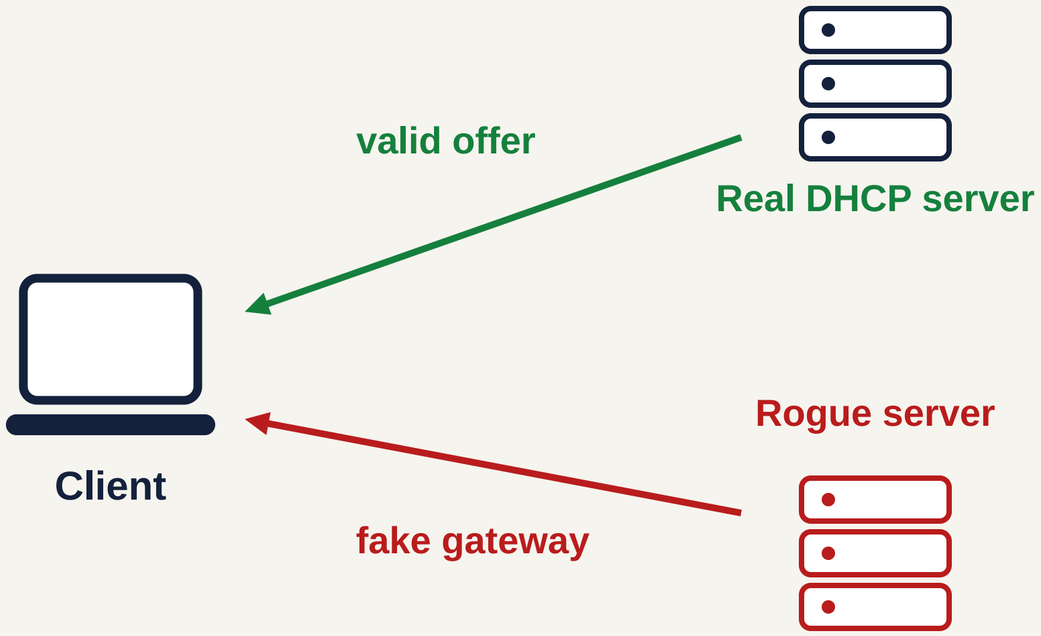
Advantages

- Less admin work
- No duplicate IP addresses
- Settings changed from one place
- Efficient reuse of addresses

Disadvantages

- Single point of failure without a backup server
- Rogue servers can mislead clients
- Starvation can exhaust the pool

Security threats and defenses



A rogue server can answer first

Threats

- Rogue DHCP server
- Starvation attack

Defenses

- DHCP snooping on switches
- Port security
- Rate limiting
- Backup DHCP servers

DHCP vs static IP vs APIPA

Feature	DHCP	Static IP	APIPA
Setup	Automatic	Manual	Automatic
Range	Any network	Any network	169.254.0.0/16
Conflicts	Avoided	Possible	Possible
Gateway, DNS	Provided	Set by hand	Not provided

Tip: an address starting with **169.254** usually means the DHCP server could not be reached.

Summary and Q&A

- DHCP automates IP configuration
- DORA: Discover, Offer, Request, Acknowledge
- Leases renew at 50% and 87.5%
- Relay agents let one server serve many subnets
- DHCP snooping protects against rogue servers

Thank you

Questions?

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